Reg. No. | | | | | | | | | | | [18MTAD201]

# SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

## COMPUTATIONAL FLUID DYNAMICS

**Time: 03 Hours** **Maximum Marks: 100**

**Instructions:**
- **i.** Answer FIVE full questions.
- **ii** Question numbers one and two are compulsory.
- **ii.** Use calculator random numbers wherever it is not given.

**1. a. i.** How do you classify a second order partial differential equation $A\frac{\partial^2 u}{\partial x^2}+B\frac{\partial^2 u}{\partial x\partial y}+C\frac{\partial^2 u}{\partial y^2}+D\frac{\partial u}{\partial x}+E\frac{\partial u}{\partial y}+Fu+G=0$ equation as elliptic, parabolic, hyperbolic. **04**

**ii.** Write a brief note on the physical process represented by hyperbolic, parabolic, elliptic, partial differential equation, the initial and boundary conditions required to solve them. **06**

**b.** Classify the following equations as linear, non-linear, quasi-linear. **10**

**i.** $\frac{\partial u}{\partial t}+a\frac{\partial u}{\partial x}=0$

**ii.** $\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}=0$

**iii.** $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$

**iv.** $\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=0$

**2. a.** Define **10**

**i.** Consistency,

**ii.** stability,

**iii.** convergence,

**iv.** amplification factor of a numerical scheme

**b.** Find the stability using Von-neumann stability analysis for linear advection equation $\frac{\partial u}{\partial t}+a\frac{\partial u}{\partial t}=0$ discretized by the following finite difference method **10**

$$\frac{u_i^{n+1}-u_i^n}{\Delta t}+a\frac{u_i^n-u_{i-1}^n}{\Delta x}=0$$

**3. a. i.** Explain the concept of Reynolds-Averaging for turbulence modeling. **05**

**ii.** Write a brief note on k-ε two equation turbulence model. **05**

**b.** Explain physical domain of dependence and numerical domain of dependence. **10**

**OR**

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4 a. Explain **10**

i. Upwind discreitzation

ii. Downwind discretization

finite difference method PDE $\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial t} = 0.$

iii. Explain the physical significance of CFL number

b. Develop a finite difference scheme which takes care of direction of wave speeds 'a' for the PDE $\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial t} = 0$ **10**

5. a. i. Define structured grids and unstructured grids. **10**

ii. Explain the types of finite volume method based on the control volume definition

b. Consider a system of Linear hyperbolic system consisting of m equations **10**

$$\frac{\partial U}{\partial t} + A\frac{\partial U}{\partial x} = 0$$

Define characteristic variables and explain the method of obtaining the solution using characteristic variables.

**OR**

6. a. Explain flux splitting for Euler Equations. **05**

b. Explain Steger and Warming scheme for Euler Equations. **15**

7. a. Derive Rankine-Hugoniot conditions for speed S of discontinuity for simplified model of scalar non-linear conservation law given by **10**

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

with initial conditions given by $u(x,0) = u_0(x)$

b. Explain the solution for the inviscid Burgers equation **10**

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0,\ f(u) = \frac{1}{2}u^2$$

$$u(x,0) = u_0(x) = \begin{cases} u_L & if \quad x < 0 \\ u_R & if \quad x > 0 \end{cases}$$

for $u_L > u_R$

**OR**

8. a. Explain Godunov scheme for solving scalar conservation law **12**

$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$ with $f = au$

b. Briefly describe HLLC Riemman solver for one dimensional Euler Equations **08**